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Support vector machine: A tool for mapping mineral prospectivity

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ABSTRACT

In this contribution, we describe an application of support vector machine (SVM), a supervised learning algorithm, to mineral prospectivity mapping. The free *R* package e1071 is used to construct a SVM with sigmoid kernel function to map prospectivity for Au deposits in western Meguma Terrain of Nova Scotia (Canada). The SVM classification accuracies of 'deposit' are 100%, and the SVM classification accuracies of the 'non-deposit' are greater than 85%. The SVM classifications of mineral prospectivity have 5–9% lower total errors, 13–14% higher false-positive errors and 25–30% lower false-negative errors compared to those of the WofE prediction. The prospective target areas predicted by both SVM and WofE reflect, nonetheless, controls of Au deposit occurrence in the study area by NE–SW trending anticlines and contact zones between Goldenville and Halifax Formations. The results of the study indicate the usefulness of SVM as a tool for predictive mapping of mineral prospectivity.

1. Introduction

Mapping of mineral prospectivity is crucial in mineral resources exploration and mining. It involves integration of information from diverse geoscience datasets including geological data (e.g., geological map), geochemical data (e.g., stream sediment geochemical data), geophysical data (e.g., magnetic data) and remote sensing data (e.g., multispectral satellite data). These sorts of data can be visualized, processed and analyzed with the support of computer and GIS techniques. Geocomputational techniques for mapping mineral prospectivity include weights of evidence (WofE) (Bonham-Carter et al., 1989), fuzzy WofE (Cheng and Agterberg, 1999), logistic regression (Agterberg and Bonham-Carter, 1999), fuzzy logic (FL) (Ping et al., 1991), evidential belief functions (EBF) (An et al., 1992; Carranza and Hale, 2003; Carranza et al., 2005), neural networks (NN) (Singer and Kouda, 1996; Porwal et al., 2003, 2004), a 'wildcat' method (Carranza, 2008, 2010; Carranza and Hale, 2002) and a hybrid method (e.g., Porwal et al., 2006; Zuo et al., 2009). These techniques have been developed to quantify indices of occurrence of mineral deposit occurrence by integrating multiple evidence layers. Some geocomputational techniques can be performed using popular software packages, such as ArcWofE (a free ArcView extension) (Kemp et al., 1999), ArcSDM 9.3 (a free ArcGIS 9.3 extension) (Sawatzky et al., 2009), MI-SDM 2.50 (a MapInfo extension) (Avantra Geosystems, 2006), GeoDAS (developed based on MapObjects, which is an Environmental Research Institute Development Kit) (Cheng, 2000). Other geocomputational techniques (e.g., FL and NN) can be performed by using *R* and Matlab.

Geocomputational techniques for mineral prospectivity mapping can be categorized generally into two types – knowledge-driven and data-driven – according to the type of inference mechanism considered (Bonham-Carter 1994; Pan and Harris 2000; Carranza 2008). Knowledge-driven techniques, such as those that apply FL and EBF, are based on expert knowledge and experience about spatial associations between mineral prospectivity criteria and mineral deposits of the type sought. On the other hand, data-driven techniques, such as WofE and NN, are based on the quantification of spatial associations between mineral prospectivity criteria and known occurrences of mineral deposits of the type sought. Additional, the mixing of knowledge-driven and data-driven methods also is used for mapping of mineral prospectivity (e.g., Porwal et al., 2006; Zuo et al., 2009). Every geocomputational technique has advantages and disadvantages, and one or the other may be more appropriate for a given geologic environment and exploration scenario (Harris et al., 2001). For example, one of the advantages of WofE is its simplicity, and straightforward interpretation of the weights (Pan and Harris, 2000), but this model ignores the effects of possible correlations amongst input predictor patterns, which generally leads to biased prospectivity maps by assuming conditional independence (Porwal et al., 2010). Comparisons between WofE and NN, NN and LR, WofE, NN and LR for mineral prospectivity mapping can be found in Singer and Kouda (1999), Harris and Pan (1999) and Harris et al. (2003), respectively.

Mapping of mineral prospectivity is a classification process, because its product (i.e., index of mineral deposit occurrence) for every location is classified as either prospective or non-prospective according to certain combinations of weighted mineral prospectivity criteria. There are two types of classification techniques.

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One type is known as supervised classification, which classifies mineral prospectivity of every location based on a training set of locations of known deposits and non-deposits and a set of evidential data layers. The other type is known as unsupervised classification, which classifies mineral prospectivity of every location based solely on feature statistics of individual evidential data layers.

A support vector machine (SVM) is a model of algorithms for supervised classification (Vapnik, 1995). Certain types of SVMs have been developed and applied successfully to text categorization, handwriting recognition, gene-function prediction, remote sensing classification and other studies (e.g., Joachims 1998; Huang et al.,2002; Cristianini and Scholkopf, 2002; Guo et al., 2005; Kavzoglu and Colkesen, 2009). An SVM performs classification by constructing an n-dimensional hyperplane in feature space that optimally separates evidential data of a predictor variable into two categories. In the parlance of SVM literature, a predictor variable is called an attribute whereas a transformed attribute that is used to define the hyperplane is called a feature. The task of choosing the most suitable representation of the target variable (e.g., mineral prospectivity) is known as feature selection. A set of features that describes one case (i.e., a row of predictor values) is called a feature vector. The feature vectors near the hyperplane are the support feature vectors. The goal of SVM modeling is to find the optimal hyperplane that separates clusters of feature vectors in such a way that feature vectors representing one category of the target variable (e.g., prospective) are on one side of the plane and feature vectors representing the other category of the target variable (e.g., non-prospective) are on the other size of the plane. A good separation is achieved by the hyperplane that has the largest distance to the neighboring data points of both categories, since in general the larger the margin the better the generalization error of the classifier. In this paper, SVM is demonstrated as an alternative tool for integrating multiple evidential variables to map mineral prospectivity.

2. Support vector machine algorithms

Support vector machines are supervised learning algorithms, which are considered as heuristic algorithms, based on statistical learning theory (Vapnik, 1995). The classical task of a SVM is binary (two-class) classification. Suppose we have a training set composed of l feature vectors $x_i \in R^n$, where i $(=1, 2, \ldots, n)$ is the number of feature vectors in training samples. The class in which each sample is identified to belong is labeled y_i, which is equal to 1 for one class or is equal to -1 for the other class (i.e. $y_i \in \{-1,1\}$) (Huang et al., 2002). If the two classes are linearly separable, then there exists a family of linear separators, also called separating hyperplanes, which satisfy the following set of equations (Kavzoglu and

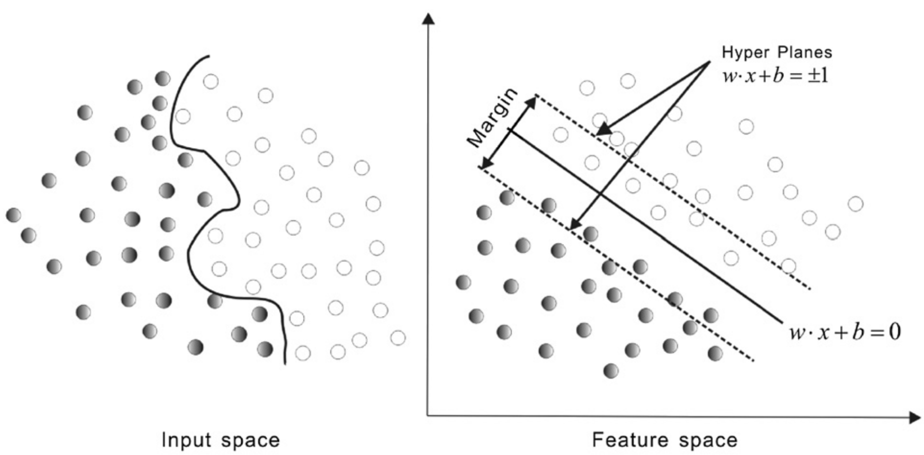

Fig. 1. Support vectors and optimum hyperplane for the binary case of linearly separable data sets.

Table 1
Experimental data.

No.	Layer A	Layer B	Layer C	Layer D	**Target**	**No.**	Layer A	Layer B	Layer C	Layer D	**Target**
1	1	1	1	1	1	21	0	0	0	0	0
2	1	1	1	1	1	22	0	0	0	0	0
3	1	1	1	1	1	23	0	0	0	0	0
4	1	1	1	1	1	24	0	1	0	0	0
5	1	1	1	1	1	25	1	0	0	0	0
6	1	1	1	1	1	26	0	0	0	0	0
7	1	1	1	1	1	27	1	1	1	0	0
8	1	1	1	1	1	28	0	0	0	0	0
9	1	1	1	0	1	29	0	0	0	0	0
10	1	1	1	0	1	30	0	0	0	0	0
11	1	0	1	1	1	31	1	1	1	0	0
12	1	1	1	0	1	32	0	0	0	0	0
13	1	1	1	0	1	33	0	0	0	0	0
14	1	1	1	0	1	34	0	0	0	0	0
15	0	1	1	0	1	35	1	0	0	0	0
16	1	0	1	0	1	36	0	0	0	0	0
17	0	1	1	0	1	37	0	0	0	0	0
18	0	1	0	1	1	38	1	1	1	0	0
19	0	1	0	1	1	29	0	0	0	0	0
20	1	0	1	0	1	40	1	0	0	0	0

Colkesen, 2009) (Fig. 1):

$$\begin{aligned} wx_i+b &\geq +1 \quad \text{for } y_i=+1 \\ wx_i+b &\leq -1 \quad \text{for } y_i=-1 \end{aligned} \tag{1}$$

which is equivalent to

$$y_i(wx_i+b) \geq 1, \quad i=1,2,\ldots,n \tag{2}$$

The separating hyperplane can then be formalized as a decision function

$$f(x)=\text{sgn}(wx+b) \tag{3}$$

where, sgn is a sign function, which is defined as follows:

$$\text{sgn}(x)=\begin{cases} 1, & \text{if } x>0 \\ 0, & \text{if } x=0 \\ -1, & \text{if } x<0 \end{cases} \tag{4}$$

The two parameters of the separating hyperplane decision function, w and b, can be obtained by solving the following optimization function:

$$\text{Minimize } \tau(w)=\frac{1}{2}\|w\|^2 \tag{5}$$

subject to

$$y_l((wx_i)+b) \geq 1, \quad i=1,\ldots,l \tag{6}$$

The solution to this optimization problem is the saddle point of the Lagrange function

$$L(w,b,\alpha)=\tfrac{1}{2}\|w\|^2-\sum_{i=1}^{l}\alpha_i(y_i((x_iw)+b)-1) \tag{7}$$

$$\frac{\partial}{\partial b}L(w,b,\alpha)=0 \quad \frac{\partial}{\partial w}L(w,b,\alpha)=0 \tag{8}$$

where α_i is a Lagrange multiplier. The Lagrange function is minimized with respect to w and b and is maximized with respect to $\alpha_i>0$. Lagrange multipliers α_i are determined by the following optimization function:

$$\text{Maximize } \sum_{i=1}^{l}\alpha_i-\tfrac{1}{2}\sum_{i,j=1}^{l}\alpha_i\alpha_jy_iy_j(x_ix_j) \tag{9}$$

subject to

$$\alpha_i \geq 0, \quad i=1,\ldots,l, \text{ and } \sum_{i=1}^{l}\alpha_iy_i=0 \tag{10}$$

The separating rule, based on the optimal hyperplane, is the following decision function:

$$f(x)=\text{sgn}\left(\sum_{i=1}^{l}y_i\alpha_i(xx_i)+b\right) \tag{11}$$

More details about SVM algorithms can be found in Vapnik (1995) and Tax and Duin (1999).

Table 2
Errors of SVM classification using linear kernel functions.

λ	Number of support vectors	Testing error (non-deposit) (%)	Testing error (deposit) (%)	Total error (%)
0.25	8	0.0	0.0	0.0
1	8	0.0	0.0	0.0
10	8	0.0	0.0	0.0
100	8	0.0	0.0	0.0
1000	8	0.0	0.0	0.0

3. Experiments with kernel functions

For spatial geocomputational analysis of mineral exploration targets, the decision function in Eq. (3) is a kernel function. The choice of a kernel function (K) and its parameters for an SVM are crucial for obtaining good results. The kernel function can be used

Table 3
Errors of SVM classification using polynomial kernel functions when $d=3$ and $r=0$.

λ	Number of support vectors	Testing error (non-deposit) (%)	Testing error (deposit) (%)	Total error (%)
0.25	12	0.0	0.0	0.0
1	6	0.0	0.0	0.0
10	6	0.0	0.0	0.0
100	6	0.0	0.0	0.0
1000	6	0.0	0.0	0.0

Table 4
Errors of SVM classification using polynomial kernel functions when $\lambda=0.25$, $r=0$.

d	Number of support vectors	Testing error (non-deposit) (%)	Testing error (deposit) (%)	Total error (%)
1	11	10.0	0.0	5.0
10	29	0.0	0.0	0.0
100	23	0.0	45.0	22.5
1000	20	0.0	90.0	45.0

Table 5
Errors of SVM classification using polynomial kernel functions when $\lambda=0.25$ and $d=3$.

r	Number of support vectors	Testing error (non-deposit) (%)	Testing error (deposit) (%)	Total error (%)
0	12	0.0	0.0	0.0
1	10	0.0	0.0	0.0
10	8	0.0	0.0	0.0
100	8	0.0	0.0	0.0
1000	8	0.0	0.0	0.0

Table 6
Errors of SVM classification using radial kernel functions.

λ	Number of support vectors	Testing error (non-deposit) (%)	Testing error (deposit) (%)	Total error (%)
0.25	14	0.0	0.0	0.0
1	13	0.0	0.0	0.0
10	13	0.0	0.0	0.0
100	13	0.0	0.0	0.0
1000	13	0.0	0.0	0.0

Table 7
Errors of SVM classification using sigmoid kernel functions when $r=0$.

λ	Number of support vectors	Testing error (non-deposit) (%)	Testing error (deposit) (%)	Total error (%)
0.25	40	0.0	0.0	0.0
1	40	0.0	35.0	17.5
10	40	0.0	6.0	3.0
100	40	0.0	6.0	3.0
1000	40	0.0	6.0	3.0

to construct a non-linear decision boundary and to avoid expensive calculation of dot products in high-dimensional feature space. The four popular kernel functions are as follows:

$$\text{Linear}: K(x_i,x_j)=\lambda x_i x_j$$
$$\text{Polynomial of degree } d: K(x_i,x_j)=(\lambda x_i x_j+r)^d, \lambda>0$$
$$\text{Radial basis function(RBF)}: K(x_i,x_j)=\exp\{-\lambda||x_i-x_j||^2\}, \lambda>0$$
$$\text{Sigmoid}: K(x_i,x_j)=\tanh(\lambda x_i x_j+r), \lambda>0 \quad (12)$$

Table 8
Errors of SVM classification using sigmoid kernel functions when $\lambda=0.25$.

r	Number of Support Vectors	Testing error (non-deposit) (%)	Testing error (deposit) (%)	Total error (%)
0	40	0.0	0.0	0.0
1	40	0.0	0.0	0.0
10	40	0.0	0.0	0.0
100	40	0.0	0.0	0.0
1000	40	0.0	0.0	0.0

The parameters λ, r and d are referred to as kernel parameters. The parameter λ serves as an inner product coefficient in the polynomial function. In the case of the RBF kernel (Eq. (12)), λ determines the RBF width. In the sigmoid kernel, λ serves as an inner product coefficient in the hyperbolic tangent function. The parameter r is used for kernels of polynomial and sigmoid types. The parameter d is the degree of a polynomial function.

We performed some experiments to explore the performance of the parameters used in a kernel function. The dataset used in the experiments (Table 1), which are derived from the study area (see below), were compiled according to the requirement for

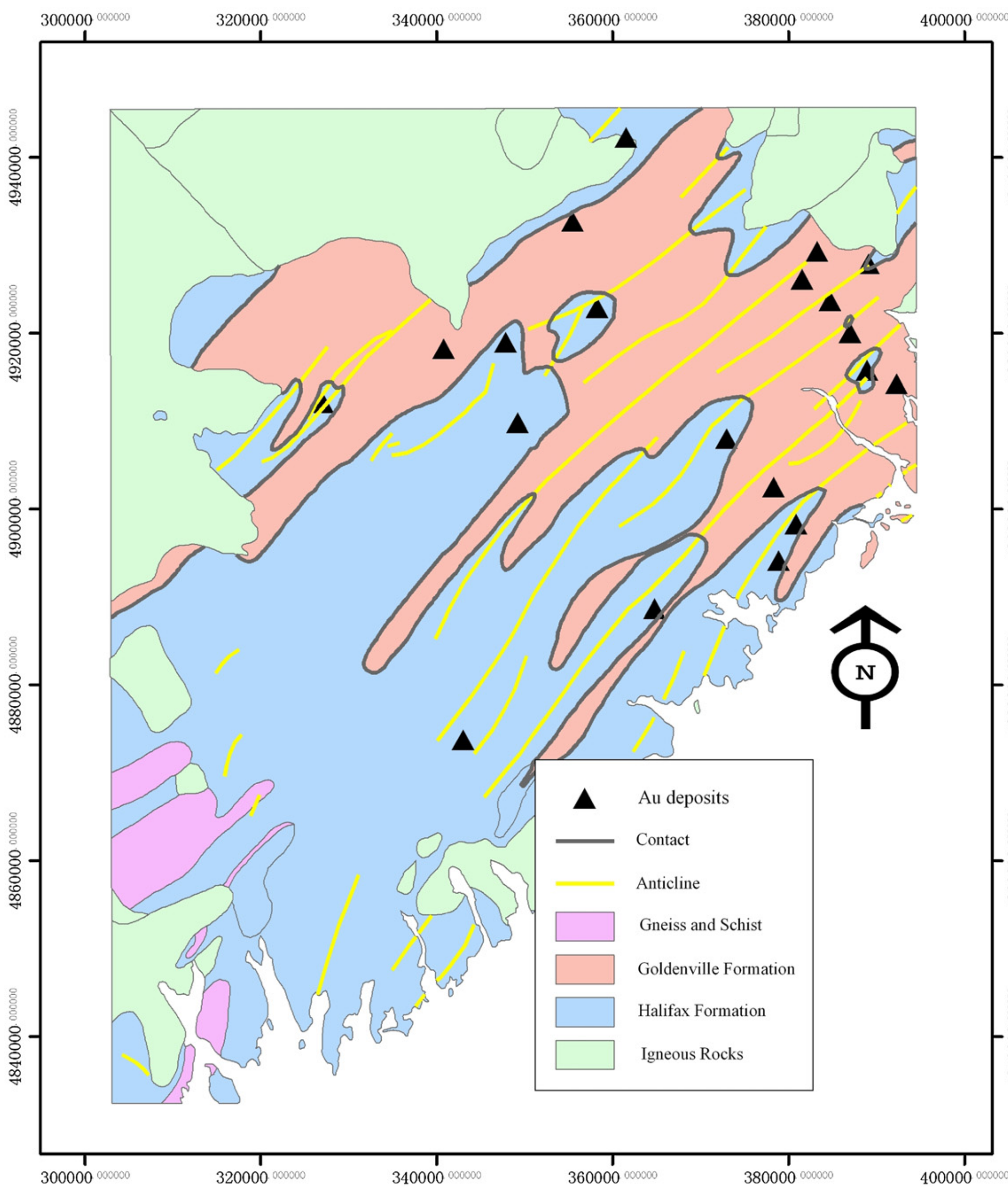

Fig. 2. Simplified geological map in western Meguma Terrain of Nova Scotia, Canada (after, Chatterjee 1983; Cheng, 2008).

classification analysis. The e1071 (Dimitriadou et al., 2010), a freeware *R* package, was used to construct a SVM. In e1071, the default values of λ, r and d are 1/(number of variables), 0 and 3, respectively. From the study area, we used 40 geological feature vectors of four geoscience variables and a target variable for classification of mineral prospectivity (Table 1). The target feature vector is either the 'non-deposit' class (or 0) or the 'deposit' class (or 1) representing whether mineral exploration target is absent or present, respectively. For 'deposit' locations, we used the 20 known Au deposits. For 'non-deposit' locations, we randomly selected them according to the following four criteria (Carranza et al., 2008): (i) non-deposit locations, in contrast to deposit locations, which tend to cluster and are thus non-random, must be random so that multivariate spatial data signatures are highly non-coherent; (ii) random non-deposit locations should be distal to any deposit location, because non-deposit locations proximal to deposit locations are likely to have similar multivariate spatial data signatures as the deposit locations and thus preclude achievement of desired results; (iii) distal and random non-deposit locations must have values for all the univariate geoscience spatial data; (iv) the number of distal and random non-deposit locations must be equal to

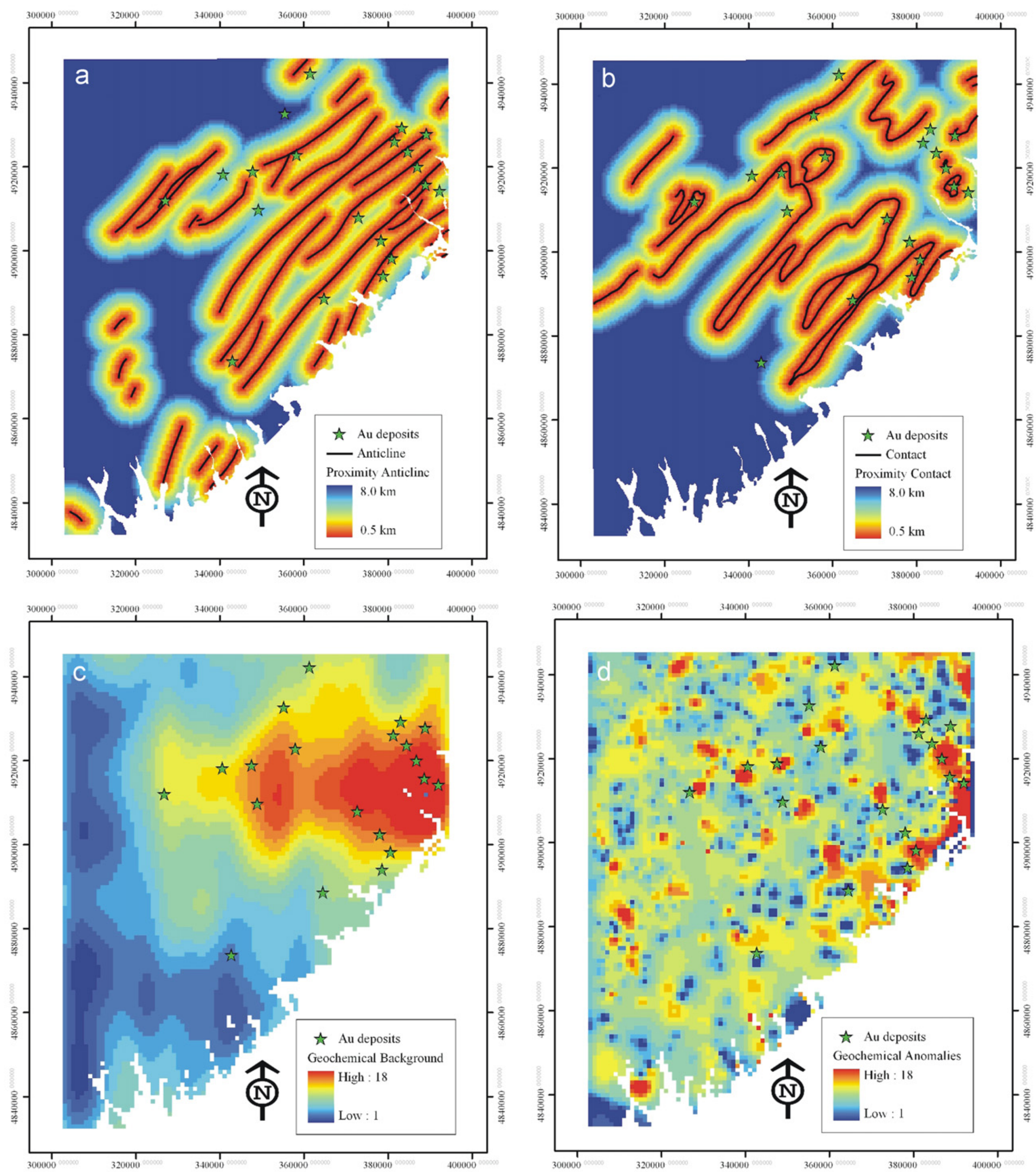

Fig. 3. Evidence layers used in mapping prospectivity for Au deposits (from Cheng, 2008): (a) and (b) represent optimum proximity to anticline axes (2.5 km) and contacts between Goldenville and Halifax formations (4 km), respectively; (c) and (d) represent, respectively, background and anomaly maps obtained via S-A filtering of the first principal component of As, Cu, Pb and Zn data.

the number of deposit locations. We used point pattern analysis (Diggle, 1983; 2003; Boots and Getis, 1988) to evaluate degrees of spatial randomness of sets of non-deposit locations and to find distance from any deposit location and corresponding probability that one deposit location is situated next to another deposit location. In the study area, we found that the farthest distance between pairs of Au deposits is 71 km, indicating that within that distance from any deposit location in there is 100% probability of another deposit location. However, few non-deposit locations can be selected beyond 71 km of the individual Au deposits in the study area. Instead, we selected random non-deposit locations beyond 11 km from any deposit location because within this distance from any deposit location there is 90% probability of another deposit location.

When using a linear kernel function and varying λ from 0.25 to 1000, the number of support vectors and the testing errors for both ‘deposit’ and ‘non-deposit’ do not vary (Table 2). In this experiment the total error of classification is 0.0%, indicating that the accuracy of classification is not sensitive to the choice of λ.

With a polynomial kernel function, we tested different values of λ, d and r as follows. If $d=3$, $r=0$ and λ is increased from 0.25 to 1000, the number of support vectors decreases from 12 to 6, but the testing errors for ‘deposit’ and ‘non-deposit’ remain nil (Table 3). If $\lambda=0.25$, $r=0$ and d is increased from 1 to 1000, the number of support vectors firstly increases from 11 to 29, then decreases from 23 to 20, the testing error for ‘non-deposit’ decreases from 10.0% to 0.0%, whereas the testing error for ‘deposit’ increases from 0.0% to 90% (Table 4). In this experiment, the total error of classification is minimum (0.0%) when $d=10$ (Table 4). If $\lambda=0.25$, $d=3$ and r is increased from 0 to 1000, the number of support vectors decreases from 12 to 8, but the testing errors for ‘deposit’ and ‘non-deposit’ remain nil (Table 5).

When using a radial kernel function and varying λ from 0.25 to 1000, the number of support vectors decreases from 14 to 13, but the testing errors of ‘deposit’ and ‘non-deposit’ remain nil (Table 6).

With a sigmoid kernel function, we experimented with different values of λ and r as follows. If $r=0$ and λ is increased from 0.25 to 1000, the number of support vectors is 40, the testing errors for ‘non-deposit’ do not change, but the testing error of ‘deposit’ increases from 0.0% to 35.0%, then decreases to 6.0% (Table 7). In this experiment, the total error of classification is minimum at 0.0% when $\lambda=0.25$ (Table 7). If $\lambda=0.25$ and r is increased from 0 to 1000, the numbers of support vectors and the testing errors of ‘deposit’ and ‘non-deposit’ do not change and the total error remains nil (Table 8).

The results of the experiments demonstrate that, for the datasets in the study area, a linear kernel function, a polynomial kernel function with $d=3$ and $r=0$, or $\lambda=0.25$, $r=0$ and $d=10$, or $\lambda=0.25$ and $d=3$, a radial kernel function, and a sigmoid kernel function with $r=0$ and $\lambda=0.25$ are optimal kernel functions. That is because the testing errors for ‘deposit’ and ‘non-deposit’ are 0% in the SVM classifications (Tables 2–8). Nevertheless, a sigmoid kernel with $\lambda=0.25$ and $r=0$, compared to all the other kernel functions, is the most optimal kernel function because it uses all the input support vectors for either ‘deposit’ or ‘non-deposit’ (Table 1) and the training and testing errors for ‘deposit’ and ‘non-deposit’ are 0% in the SVM classification (Tables 7 and 8).

4. Prospectivity mapping in the study area

The study area is located in western Meguma Terrain of Nova Scotia, Canada. It measures about 7780 km^2. The host rock of Au deposits in this area consists of Cambro-Ordovician low-middle grade metamorphosed sedimentary rocks and a suite of Devonian aluminous granitoid intrusions (Sangster, 1990; Ryan and Ramsay, 1997). The metamorphosed sedimentary strata of the Meguma Group are the lower sand-dominated flysch Goldenville Formation and the upper shaly flysch Halifax Formation occurring in the central part of the study area. The igneous rocks occur mostly in the northern part of the study area (Fig. 2).

In this area, 20 turbidite-hosted Au deposits and occurrences (Ryan and Ramsay, 1997) are found in the Meguma Group, especially near the contact zones between Goldenville and Halifax Formations (Chatterjee, 1983). The major Au mineralization-related geological features are the contact zones between Goldenville and Halifax Formations, NE–SW trending anticline axes and NE–SW trending shear zones (Sangster, 1990; Ryan and Ramsay, 1997). This dataset has been used to test many mineral prospectivity mapping algorithms (e.g., Agterberg, 1989; Cheng, 2008). More details about the geological settings and datasets in this area can be found in Xu and Cheng (2001).

We used four evidence layers (Fig. 3) derived and used by Cheng (2008) for mapping prospectivity for Au deposits in the area. Layers A and B represent optimum proximity to anticline axes (2.5 km) and optimum proximity to contacts between Goldenville and Halifax Formations (4 km), respectively. Layers C and D represent variations in geochemical background and anomaly, respectively, as modeled by multifractal filter mapping of the first principal component of As, Cu, Pb, and Zn data. Details of how the four evidence layers were obtained can be found in Cheng (2008).

4.1. Training dataset

The application of SVM requires two subsets of training locations: one training subset of ‘deposit’ locations representing presence of mineral deposits, and a training subset of ‘non-deposit’ locations representing absence of mineral deposits. The value of y_i is 1 for ‘deposits’ and -1 for ‘non-deposits’. For ‘deposit’ locations, we used the 20 known Au deposits (the sixth column of Table 1). For ‘non-deposit’ locations (last column of Table 1), we obtained two ‘non-deposit’ datasets (Tables 9 and 10) according to the above-described selection criteria (Carranza et al., 2008). We combined the ‘deposits’ dataset with each of the two ‘non-deposit’ datasets to obtain two training datasets. Each training dataset commonly contains 20 known Au deposits but contains different 20 randomly selected non-deposits (Fig. 4).

4.2. Application of SVM

By using the software e1071, separate SVMs both with sigmoid kernel with $\lambda=0.25$ and $r=0$ were constructed using the two

Table 9
The value of each evidence layer occurring in ‘non-deposit’ dataset 1.

No.	Layer A	Layer B	Layer C	Layer D
1	0	0	0	0
2	0	0	0	0
3	1	1	1	0
4	0	0	0	0
5	0	0	0	0
6	1	0	0	0
7	0	0	0	0
8	0	0	0	0
9	0	1	0	0
10	0	1	0	0
11	0	0	0	0
12	0	0	0	0
13	0	0	0	0
14	0	0	0	0
15	0	0	0	0
16	0	1	0	0
17	0	0	0	0
18	0	0	0	0
19	0	1	0	0
20	0	0	0	0

training datasets. With training dataset 1, the classification accuracies for 'non-deposits' and 'deposits' are 95% and 100%, respectively; With training dataset 2, the classification accuracies for 'non-deposits' and 'deposits' are 85% and 100%, respectively.

Table 10
The value of each evidence layer occurring in 'non-deposit' dataset 2.

No.	Layer A	Layer B	Layer C	Layer D
1	1	0	1	0
2	0	0	0	0
3	0	0	0	0
4	1	1	1	0
5	0	0	0	0
6	0	1	1	0
7	1	0	1	0
8	0	0	0	0
9	1	0	0	0
10	1	1	1	0
11	1	0	0	0
12	0	0	1	0
13	1	0	0	0
14	0	0	0	0
15	0	0	0	0
16	1	0	0	0
17	1	0	0	0
18	0	0	1	0
19	0	0	1	0
20	0	0	0	0

The total classification accuracies using the two training datasets are 97.5% and 92.5%, respectively. The patterns of the predicted prospective target areas for Au deposits (Fig. 5) are defined mainly by proximity to NE–SW trending anticlines and proximity to contact zones between Goldenville and Halifax Formations. This indicates that 'geology' is better than 'geochemistry' as evidence of prospectivity for Au deposits in this area.

With training dataset 1, the predicted prospective target areas occupy 32.6% of the study area and contain 100% of the known Au deposits (Fig. 5a). With training dataset 2, the predicted prospective target areas occupy 33.3% of the study area and contain 95.0% of the known Au deposits (Fig. 5b). In contrast, using the same datasets, the prospective target areas predicted via WofE occupy 19.3% of study area and contain 70.0% of the known Au deposits (Cheng, 2008).

The error matrices for two SVM classifications show that the type 1 (false-positive) and type 2 (false-negative) errors based on training dataset 1 (Table 11) and training dataset 2 (Table 12) are 32.6% and 0%, and 33.3% and 5%, respectively. The total errors for two SVM classifications are 16.3% and 19.15% based on training datasets 1 and 2, respectively. In contrast, the type 1 and type 2 errors for the WofE prediction are 19.3% and 30% (Table 13), respectively, and the total error for the WofE prediction is 24.65%.

The results show that the total errors of the SVM classifications are 5–9% lower than the total error of the WofE prediction. The 13–14% higher false-positive errors of the SVM classifications compared to that of the WofE prediction suggest that the SVM

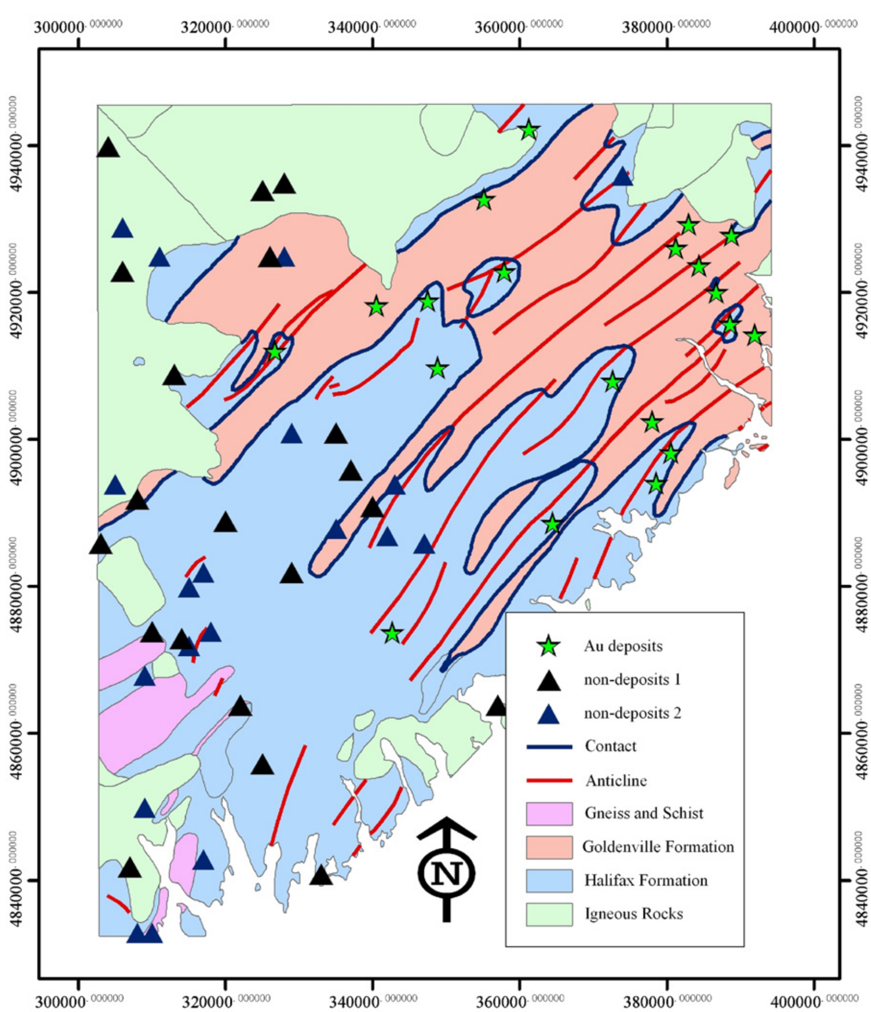

Fig. 4. The locations of 'deposit' and 'non-deposit'.

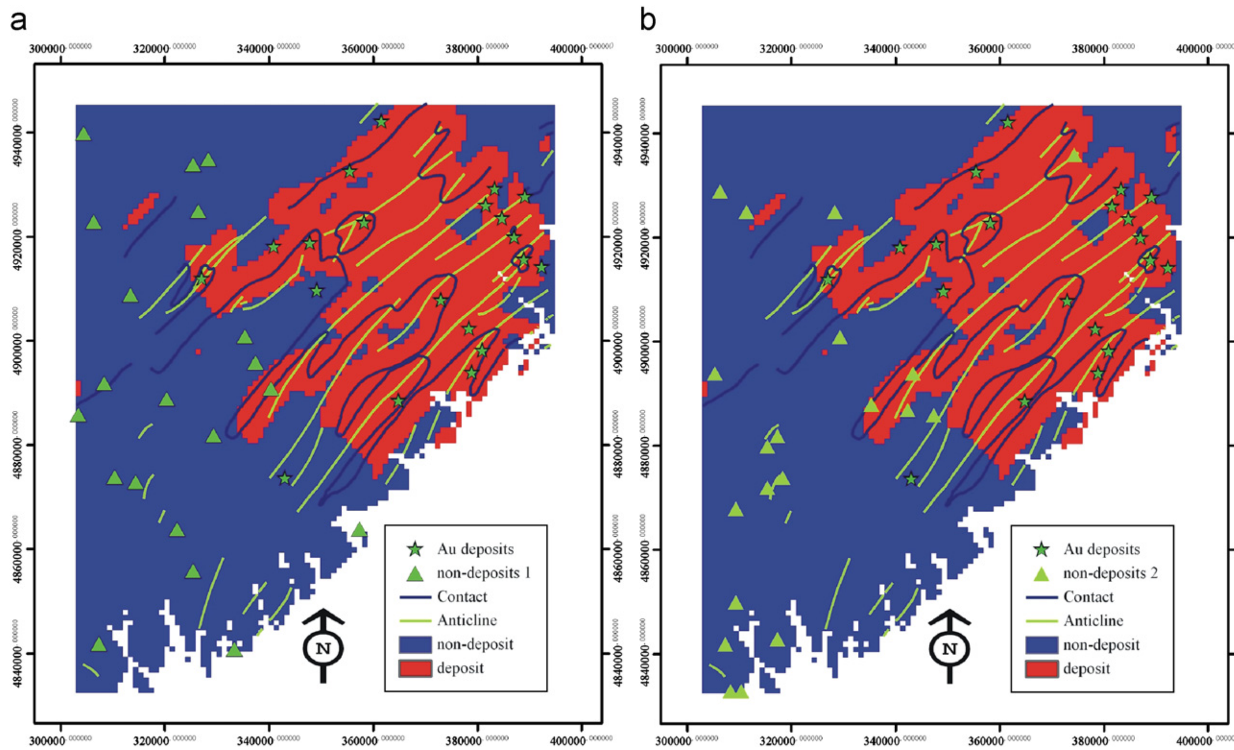

Fig. 5. Prospective targets area for Au deposits delineated by SVM. (a) and (b) are obtained using training dataset 1 and 2, respectively.

Table 11
Error matrix for SVM classification using training dataset 1.

	Known		
	All 'deposits'	All 'non-deposits'	Total
Prediction			
'Deposit'	**100**	**32.6**	132.6
'Non-deposit'	**0**	**67.4**	67.4
Total	100	100	200

Type 1 (false-positive) error = **32.6**.
Type 2 (false-negative) error = **0**.
Total error = **16.3**.
Note: Values in the matrix are percentages of 'deposit' and 'non-deposit' locations.

Table 12
Error matrix for SVM classification using training dataset 2.

	Known		
	All 'deposits'	All 'non-deposits'	Total
Prediction			
'Deposits'	**95**	**33.3**	128.3
'Non-deposits'	**5**	**66.7**	71.4
Total	100	100	200

Type 1 (false-positive) error = 33.3.
Type 2 (false-negative) error = 5.
Total error = 19.15.
Note: Values in the matrix are percentages of 'deposit' and 'non-deposit' locations.

classifications result in larger prospective areas that may not contain undiscovered deposits. However, the 25–30% higher false-negative error of the WofE prediction compared to those of the SVM classifications suggest that the WofE analysis results in larger non-prospective areas that may contain undiscovered deposits. Certainly, in mineral exploration the intentions are not

Table 13
Error matrix for WofE prediction.

	Known		
	All 'deposits'	All 'non-deposits'	Total
Prediction			
'Deposit'	**70**	**19.3**	89.3
'Non-deposit'	**30**	**80.7**	110.7
Total	100	100	200

Type 1 (false-positive) error = 19.3.
Type 2 (false-negative) error = 30.
Total error = 24.65.
Note: Values in the matrix are percentages of 'deposit' and 'non-deposit' locations.

to miss undiscovered deposits (i.e., avoid false-negative error) and to minimize exploration cost in areas that may not really contain undiscovered deposits (i.e., keep false-positive error as low as possible). Thus, results suggest the superiority of the SVM classifications over the WofE prediction.

5. Conclusions

Nowadays, SVMs have become a popular geocomputational tool for spatial analysis. In this paper, we used an SVM algorithm to integrate multiple variables for mineral prospectivity mapping. The results obtained by two SVM applications demonstrate that prospective target areas for Au deposits are defined mainly by proximity to NE–SW trending anticlines and to contact zones between the Goldenville and Halifax Formations. In the study area, the SVM classifications of mineral prospectivity have 5–9% lower total errors, 13–14% higher false-positive errors and 25–30% lower false-negative errors compared to those of the WofE prediction. These results indicate that SVM is a potentially useful tool for integrating multiple evidence layers in mineral prospectivity mapping.

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References

Agterberg, F.P., 1989. Computer programs for mineral exploration. Science 245, 76–81.

Agterberg, F.P., Bonham-Carter, G.F., 1999. Logistic regression and weights of evidence modeling in mineral exploration. In: Proceedings of the 28th International Symposium on Applications of Computer in the Mineral Industry (APCOM), Golden, Colorado, pp. 483–490.

An, P., Moon, W.M., Bonham-Carter, G.F., 1992. On knowledge-based approach on integrating remote sensing, geophysical and geological information. In: Proceedings of International Geoscience and Remote Sensing Symposium (IGARSS), 1992, pp. 34–38.

Avantra Geosystems, 2006. A Tutorial Guide to using MI-SDM v2.50 based on USGS Open-File Report 01-221 by Gary L. Raines.

Bonham-Carter, G.F., 1994. Geographic Information Systems for Geoscientists: Modelling with GIS. Pergamon, Ontario, 398 pp.

Bonham-Carter, G.F., Agterberg, F.P., Wright, D.F., 1989. Weights of evidence modelling: a new approach to mapping mineral potential. In: Agterberg, F.P., Bonham-Carter, G.F. (Eds.), Statistical Applications in the Earth Sciences. Geological Survey of Canada, Paper 89-9, pp. 171–183.

Boots, B.N., Getis, A., 1988. Point Pattern Analysis. Sage University Scientific Geography, Series No. 8. Sage Publications, Beverly Hills, 93 pp.

Carranza, E.J.M., 2008. Geochemical anomaly and mineral prospectivity mapping in GIS. In: Handbook of Exploration and Environmental Geochemistry, vol. 11, Elsevier, Amsterdam, 351 pp.

Carranza, E.J.M., 2010. Improved wildcat modelling of mineral prospectivity. Resource Geology 60, 129–149.

Carranza, E.J.M., Hale, M., 2002. Wildcat mapping of gold potential, Baguio district, Philippines. Transactions of the Institution of Mining and Metallurgy (Section B—Applied Earth Science) 111, 100–105.

Carranza, E.J.M., Hale, M., 2003. Evidential belief functions for geologically constrained mapping of gold potential, Baguio district, Philippines. Ore Geology Reviews 22, 117–132.

Carranza, E.J.M., Hale, M., Faassen, C., 2008. Selection of coherent deposit-type locations and their application in data-driven mineral prospectivity mapping. Ore Geology Reviews 33, 536–558.

Carranza, E.J.M., Woldai, T., Chikambwe, E.M., 2005. Application of data-driven evidential belief functions to prospectivity mapping for aquamarine-bearing pegmatites, Lundazi District, Zambia. Natural Resources Research 14, 47–63.

Chatterjee, A.K., 1983. Metallogenic map of Nova Scotia, version 1, scale 1:500 000. Department of Mines and Energy, Nova Scotia, Canada.

Cheng, Q., 2000. GeoData Analysis System (GeoDAS) for mineral Exploration: User's Guide and Exercise Manual. Material for the training workshop on GeoDAS held at York University, Toronto, Canada, 1, 3, 204, 〈http://www.gisworld.org/geodas〉.

Cheng, Q., 2008. Non-linear theory and power-law models for information integration and mineral resources quantitative assessments. Mathematical Geosciences 40, 503–532.

Cheng, Q., Agterberg, F.P., 1999. Fuzzy weights of evidence and its application in mineral potential mapping. Natural Resources Research 8, 27–35.

Cristianini, N., Scholkopf, B., 2002. Support vector machines and kernel methods—the new generation of learning machines. AI Magazine 23, 31–41.

Diggle, P.J., 1983. Statistical Analysis of Spatial Point Patterns. Academic Press, London, 148 pp.

Diggle, P.J., 2003. Statistical Analysis of Spatial Point Patterns 2nd edn. Arnold, London, UK.

Dimitriadou, E., Hornik, K., Leisch, F., Meyer, D., Weingessel, A., 2010. e1071: Miscellaneous Functions of the Department of Statistics (e1071), TU Wien. 〈http://cran.r-project.org/web/packages/e1071/index.html〉.

Guo, Q.H., Kelly, M., Graham, C.H., 2005. Support vector machines for prediction distribution of Sudden Oak Death in California. Ecological Modeling 182, 75–90.

Harris, D.P., Pan, G., 1999. Mineral favorability mapping: a comparison of artificial neural networks, logistic regression and discriminant analysis. Natural Resources Research 8, 93–109.

Harris, D.P., Zurcher, L., Stanley, M., Marlow, J., Pan, G., 2003. A comparative analysis of favourability mappings by weights of evidence, probabilistic neural networks, discriminant analysis, and logistic regression. Natural Resources Research 12, 241–255.

Harris, J.R., Wilkinson, L., Heather, K., Fumerton, S., Bernier, M.A., Ayer, J., Dahn, R., 2001. Application of GIS processing techniques for producing mineral prospectivity maps—a case study: mesothermal Au in the Swayze Greenstone Belt, Ontario, Canada. Natural Resources Research 10, 91–124.

Huang, C., Davis, L.S., Townshend, J.R.G., 2002. An assessment of support vector machines for land cover classification. International Journal of Remote Sensing 23, 725–749.

Joachims, T., 1998. Text categorization with support vector machines: learning with many relevant features. In: Proceedings of European Conference on Machine Learning, Springer-Verlag, Berlin, pp. 137–142.

Kavzoglu, T., Colkesen, I., 2009. A kernel functions analysis for support vector machines for land cover classification. International Journal of Applied Earth Observation and Geoinformation 11, 352–359.

Kemp, L.D., Bonham-Carter, G.F., Raines, G.L., 1999. ArcWofE: ArcView extension for weights of evidence mapping: 〈http://gis.nrcan.gc.ca/software/arcview/wofe〉.

Pan, G., Harris, D.P., 2000. Information Synthesis for Mineral Exploration. Oxford Univ. Press, New York, 461 pp.

Ping, A., Wooil, M.M., Andy, R., 1991. Application of fuzzy set theory to integrated mineral exploration. Canada of Journal of Exploration Geophysics 27, 1–11.

Porwal, A., Carranza, E.J.M., Hale, M., 2003. Artificial neural networks for mineral-potential mapping: a case study from Aravalli Province, Western India. Natural Resources Research 12, 156–171.

Porwal, A., Carranza, E.J.M., Hale, M., 2004. A hybrid neuro-fuzzy model for mineral potential mapping. Mathematical Geology 36, 803–826.

Porwal, A., Carranza, E.J.M., Hale, M., 2006. A hybrid fuzzy weights-of-evidence model for mineral potential mapping. Natural Resources Research 15, 1–14.

Porwal, A., González-Álvarez, I., Markwitz, V., McCuaig, T.C., Mamuse, A.A., 2010. Weights of evidence and logistic regression modeling of magmatic nickel sulfide prospectivity in the Yilgarn Craton, Western Australia, Ore Geology Reviews 38, 184–196.

Ryan, R.J., Ramsay, W.R.H., 1997. Preliminary comparison of gold field in the Meguma Terrain, Nova Scotia, and Victoria, Australia. In: Macdonald, D.R., Mills, K.A. (Eds.), Mines and Mineral Branch. Report of Activities 1996, 97-1, pp. 157–162.

Sangster, A.L. 1990. Metallogeny of the Meguma Terrain, Nova Scotia. In: Sangster, A.L. (Ed.), Mineral Deposit Studies In Nova Scotia. Geological Survey of Canada. Paper, 90-8(1), pp. 115–152.

Sawatzky, D.I., Raines, G.L., Bonham-Carter, G.F., Looney, 2009. Spatial Data Modeller (SDM): ArcMap 9.3 geoprocessing tools for spatial modeling using weights of evidences, logistic regression, fuzzy logic and neural networks. 〈http://arcscripts.esri.com/details.asp/dbid=15341〉.

Singer, D.A., Kouda, R., 1996. Application of a feedforward neural network in the search for Kuruko deposits in the Hokuroku district, Japan. Mathematical Geology 28, 1017–1023.

Singer, D.A., Kouda, R., 1999. A comparison of the weights-of evidence method and probabilistic neural networks. Natural Resources Research 8, 287–298.

Tax, D., Duin, E., 1999. Support vector domain description. Pattern Recognition Letters 20, 1191–1199.

Vapnik, V., 1995. Nature of Statistical Learning Theory. John Wiley and Sons, Inc., New York.

Xu, Y., Cheng, Q., 2001. A fractal filtering technique for processing regional geochemical maps for mineral exploration. Geochemistry: Exploration, Environment Analysis 1, 147–156.

Zuo, R., Cheng, Q., Agterberg, F.P., 2009. Application of a hybrid method combining multilevel fuzzy comprehensive evaluation with asymmetric fuzzy relation analysis to mapping prospectivity. Ore Geology Reviews 35, 101–108.